

ESG Momentum Radar Data Workflow Report

Prepared for frontend handoff | Prototype data layer | Time window: 2025-06-10 to 2026-06-10

Summary

We built the ESG Momentum Radar data workflow up to the structured dataset stage for the hackathon prototype. The workflow starts from a manually curated list of Singapore-focused companies and source URLs, uses Firecrawl for live one-time collection where pages can be scraped reliably, stores the collected text in a raw source JSON database, filters the raw text into evidence-backed ESG signals across Environmental, Social, and Governance, then aggregates those signals into company-level datasets and quarterly evidence-trend outputs for the future frontend. The pipeline also includes validation and a QA feedback loop to catch hidden consistency issues such as missing ESG coverage, invalid source references, non-matching evidence quotes, wrong time windows, and trend or dataset totals that do not match the underlying signal rows. The frontend has not been implemented yet; these JSON files are the current data endpoints to consume.

Pipeline Commands

Command	Purpose
npm run collect	Runs Firecrawl/live collection or transparent seed fallback into raw_sources.json.
npm run extract	Filters raw source text into row-level ESG signals.
npm run dataset	Builds company-level structured ESG dataset for dashboard use.
npm run trends	Builds quarterly evidence trend points.
npm run pipeline	Runs collect -> extract -> dataset -> trends.
npm run validate	Validates references, time windows, evidence quotes, and output shape.
npm run qa	Runs the deeper feedback-loop QA checks.
npm run check	Runs TypeScript type checking.

Frontend Data Endpoints

JSON endpoint	Frontend use	Important keys
data/structured_esg_dataset.json	Main dashboard data: company cards, ranking table, E/S/G breakdown, summary metrics, top evidence.	company_id, company_name, sector, sgx_identifier, time_window, total_signal_count, total_signal_score, average_confidence, source_count, source_platform_counts, category_breakdown, top_evidence, dataset_disclaimer
data/structured_esg_signals.json	Evidence panel and drilldowns: individual ESG signals with source-backed quotes.	signal_id, company_id, source_id, esg_category, signal_tags, signal_direction, signal_strength, confidence, time_relevance, evidence_summary, evidence_quote, source_reliability, weighted_signal_score, published_date, time_window, source_platform, url, prototype_disclaimer
data/trend_output.json	Trend charts: quarterly evidence signal movement by company and E/S/G category.	company_id, company_name, sector, period, period_start, period_end, environmental_score, social_score, governance_score, total_signal_score, environmental_signal_count, social_signal_count, governance_signal_count, positive_signal_count, negative_signal_count, confidence, source_count, trend_disclaimer

JSON endpoint	Frontend use	Important keys
data/raw_sources.json	Source provenance and debugging: raw scraped or fallback text.	source_id, seed_id, company_id, source_platform, url, title, source_type, scraped_at, raw_text, extraction_status, content_hash, source_reliability, published_date, collection_window, firecrawl_metadata
data/companies.json	Company metadata and labels.	company_id, name, sector, sgx_identifier, website, initial_esg_score, historical_esg_scores, prototype_note
data/collection_scope.json	Display scope, collection window, source priorities, and ESG dimension definitions.	window_start, window_end, window_label, companies, platform_priority, esg_dimensions

Environment Keys

Key	Current role	Frontend exposure
FIRECRAWL_API_KEY	Used by the local/backend collection script when COLLECTION_MODE=live. It must stay in .env.	Do not expose in frontend.
COLLECTION_MODE	live means the collector calls Firecrawl; mock uses seeded prototype text.	Not needed in frontend.
AI_EXTRACTION_MODE	mock currently; extraction is rules-based. live would require OPENAI_API_KEY.	Not needed in frontend.
OPENAI_API_KEY	Currently blank/not required. Only needed if AI_EXTRACTION_MODE=live later.	Do not expose in frontend.
OPENAI_MODEL	Default model name for future live AI extraction.	Not needed in frontend.

Current QA Status

Check	Latest result
Companies	10
Seeds / raw sources	40 / 40
Structured ESG signals	75
Structured company dataset rows	10
Trend points	21
Core ESG categories	Environmental: 26, Social: 19, Governance: 30
Raw source statuses	skipped_unchanged=18, fallback_seeded=22
QA feedback loop	0 errors, 0 warnings
Security audit	0 vulnerabilities

Frontend Consumption Recommendation

For the first frontend version, use structured_esg_dataset.json for the dashboard overview and ranking table, structured_esg_signals.json for evidence panels and company drilldowns, and trend_output.json for quarterly trend visuals. The Firecrawl key and any future OpenAI key must remain server-side or local-only; the frontend should consume generated JSON data, not call external scraping or AI services directly.